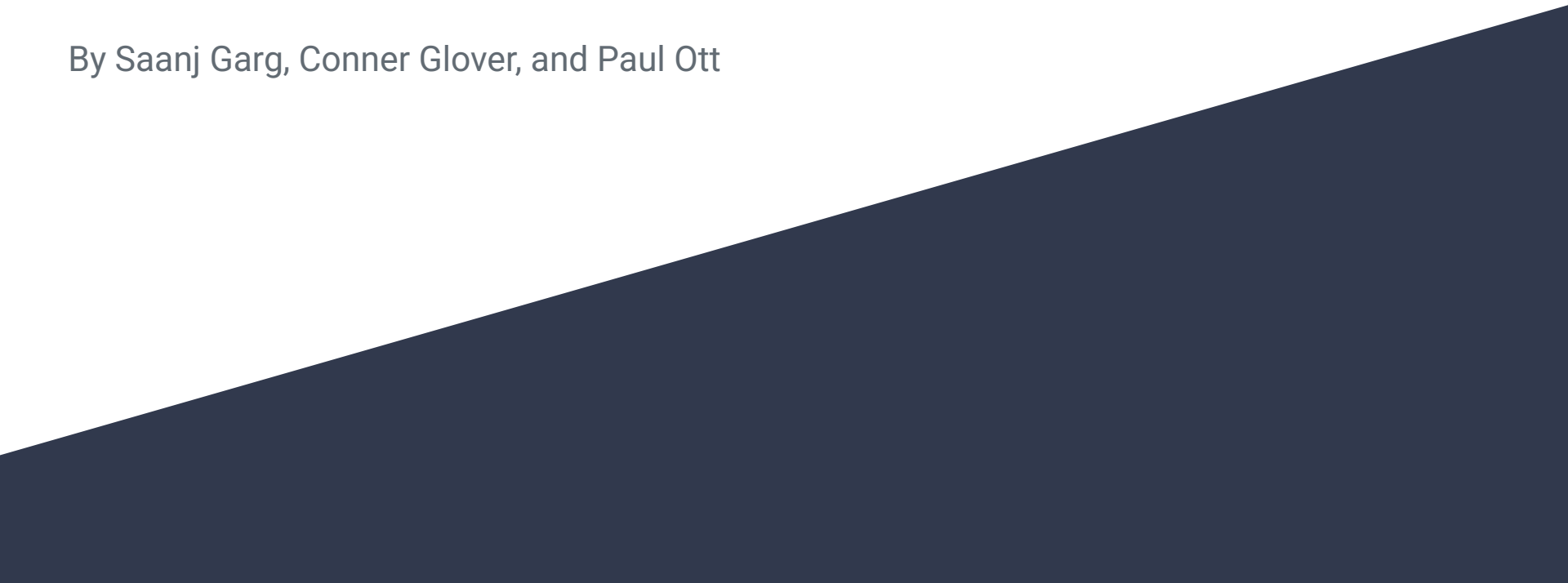


1.1.6

Compound Machine

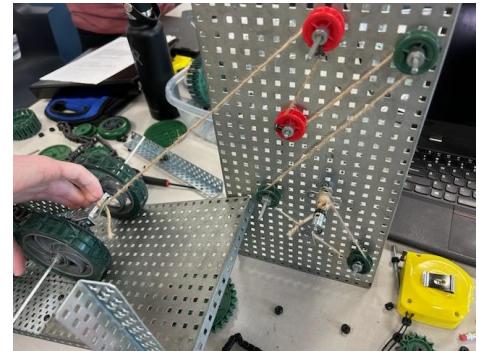
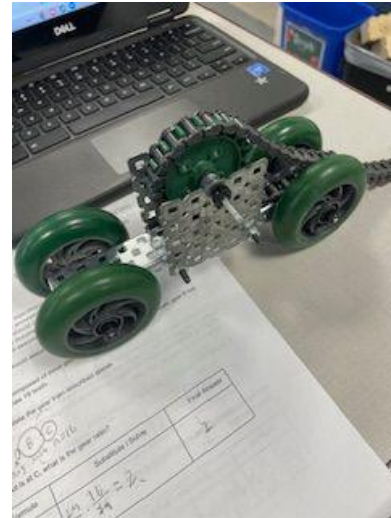
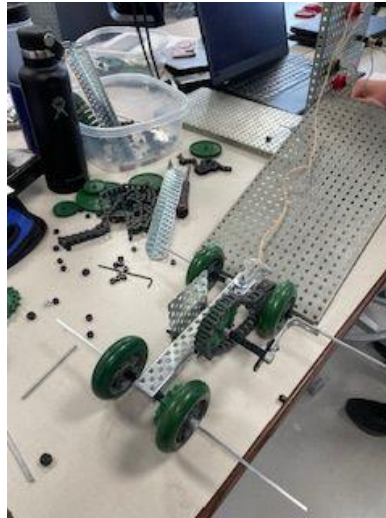
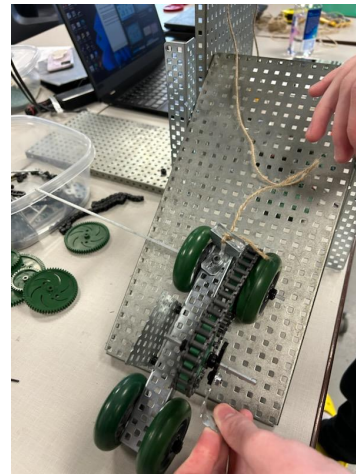
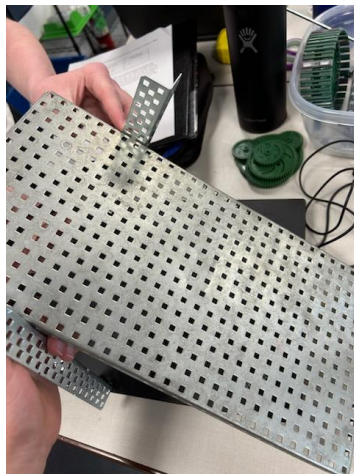
By Saanj Garg, Conner Glover, and Paul Ott

A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

Process & Brainstorming

We started out with the idea to use a pulley to pull a car up a ramp. After starting to build, we changed the car's design by switching out wheels and adding gears. This way, the car could more easily drive up the ramp. We added strands to the pulley to make it easier to pull the car. We also then turned this into a game where you're fighting against a resistance created by another person. We created the pulley by adding many smooth gears for the string to be pulled along. We then added a crank to pull the string so the car can go up the ramp.

Pictures



Pictures (*Cont.*)



Videos



Math

- Wheel:
 - Small: 2.5", Big: 5"
 - Axle 0.12"(for both)
 - Small wheel IMA: 20.8
 - Big wheel IMA: 41.7
- Ramp:
 - A:13.9553", B:5.5", C:16"
 - Angle: 21.5102°
 - IMA: 2.9
- Gears:
 - Input: 24T
 - Output: 16T
 - Gear Ratio: 3:2
 - IMA:1.5
- Pulley
 - Number of strands: 5
 - IMA: 5
- Final IMA: 21.75

R/W/L

Right	Wrong	Learned
• Made an elaborate project. • Gave it a function rather than just making a useless compound system. • Used tape labels to make it more interactive. • Finished on time. • Cleaned up all extra parts. • Made a good presentation.	• Game is kind of unfair. • It's a bit difficult to fight against if you're on the green team because of friction knotting. • If you start out with too much slack, green will win immediately • The design of the car was difficult to get right.	• How to incorporate different design elements to make one system. • How to make a system with a purpose. • How to add entertainment to something otherwise boring. • How to think of another solution to the design if it wasn't working.